
Forecasting dengan Earn Value Analysis dan Metode Statistik

Perbandingan antara Teori Earn Value Analysis (EVA) dengan Exponential Smoothing Holt's Linear Damped Trend Model dan Logistic Growth Model Non Linear Square pada Data Proyek

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DATA AWAL PROYEK

Total Penjualan : Rp 59.233.648.767
Rencana Budget : Rp 50.568.600.331

Bulan	Progress Penjualan	Biaya
Maret	Rp 31.243.672.753	Rp 26.131.324.832
April	Rp 35.440.897.788	Rp 32.858.462.674
Mei	Rp 38.580.386.013	Rp 36.574.246.756
Juni	Rp 47.988.586.039	Rp 44.975.149.464
Juli	Rp 52.872.221.393	Rp 49.614.523.315
Agustus	Rp 52.872.221.393	Rp 49.676.829.516
September	Rp 52.872.221.393	Rp 49.837.200.480
Oktober	Rp 54.254.144.494	Rp 50.535.677.086
November	Rp 54.254.144.494	Rp 50.535.677.086
Desember	Rp 54.463.040.172	Rp 50.746.497.979
Januari	Rp 54.463.040.172	Rp 50.942.464.827
Februari	Rp 54.944.673.223	Rp 51.058.673.197
Maret	Rp 55.093.513.628	Rp 51.190.856.981
April	Rp 57.827.183.059	Rp 52.249.072.088
Mei	Rp 58.397.633.059	Rp 52.249.072.088



Dasar Teori



Earned Value Analysis

Earned Value Analysis (EVA) merupakan metode manajemen proyek yang menggabungkan ruang lingkup, waktu, biaya untuk menilai kemajuan suatu proyek

Rumus-rumus:

- Cost Variance = $EV - AC$
- Schedule Variance = $EV - PV$
- Cost Performance Index = EV/AC
- Schedule Performance Index = EV/PV
- Estimate to Complete = $(BAC - EV) / CPI$
- Estimate at Completion = $ETC + AC$

Holt's Damped Trend

Exponential Smoothing Holt's Damped Trend merupakan teknik forecasting berdasarkan kebiasaan naik atau turunnya data historis yang memiliki parameter phi, yang membuat laju kenaikannya makin lama makin lambat

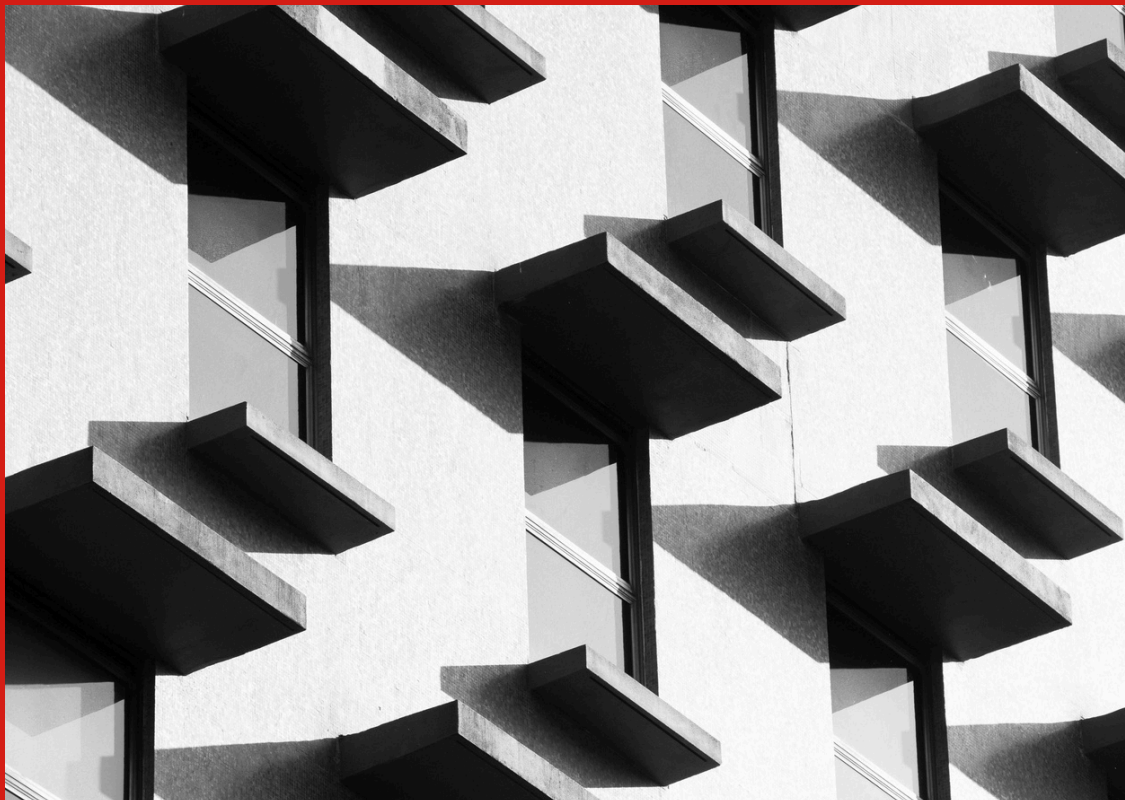
- alfa(α) = smoothing level → mengontrol seberapa cepat level dalam merespon data baru
- beta(β) = smoothing trend → mengontrol seberapa cepat tren atau kemiringan dalam menyesuaikan level
- Phi (ϕ) = parameter damping → ngontrol seberapa cepat level meredam atau damping trend

Logistic Curve Model

kurva berbentuk s dan polanya naik cepat di awal, kemudian melambat dan mendekati suatu nilai batas maksimal (K)

- $P(0)$ = Nilai awal pada $t=0$
- K = batas maksimum (carrying capacity)
- r = laju pertumbuhan intrinsik → seberapa cepat pertumbuhan di fase awal

Earn Value Analysis (EVA)



MAPE
5,4%

Actual Cost	CPI	Earn Value (EV)	ETC	EAC
Biaya	EV/AC	Progress Penjualan	(TP-EV)/CPI	ETC + AC
Rp 26.131.324.832	1,119	Rp 31.243.672.753	Rp 23.410.024.841	Rp 49.541.349.674
Rp 32.858.462.674	1,078	Rp 35.440.897.788	Rp 22.059.069.289	Rp 54.917.531.963
Rp 36.574.246.756	1,055	Rp 38.580.386.013	Rp 19.579.314.940	Rp 56.153.561.696
Rp 44.975.149.464	1,067	Rp 47.988.586.039	Rp 10.538.930.581	Rp 55.514.080.045
Rp 49.614.523.315	1,066	Rp 52.872.221.393	Rp 5.969.470.895	Rp 55.583.994.211
Rp 49.676.829.516	1,064	Rp 52.872.221.393	Rp 5.976.967.391	Rp 55.653.796.907
Rp 49.837.200.480	1,061	Rp 52.872.221.393	Rp 5.996.262.745	Rp 55.833.463.224
Rp 50.535.677.086	1,073	Rp 54.254.144.494	Rp 4.638.219.299	Rp 55.173.896.385
Rp 50.535.677.086	1,073	Rp 54.254.144.494	Rp 4.638.219.299	Rp 55.173.896.385
Rp 50.746.497.979	1,073	Rp 54.463.040.172	Rp 4.445.063.637	Rp 55.191.561.615
Rp 50.942.464.827	1,069	Rp 54.463.040.172	Rp 4.462.229.060	Rp 55.404.693.887
Rp 51.058.673.197	1,076	Rp 54.944.673.223	Rp 3.985.634.781	Rp 55.044.307.978
Rp 51.190.856.981	1,076	Rp 55.093.513.628	Rp 3.846.860.580	Rp 55.037.717.561
Rp 52.249.072.088	1,107	Rp 57.827.183.059	Rp 1.270.795.572	Rp 53.519.867.661
Rp 52.249.072.088	1,118	Rp 58.397.633.059	Rp 747.993.415	Rp 52.997.065.504

Exponential Smoothing Holt's Damped Trend

Parameter		Bulan	Biaya	t	Xt	Level (L)	Tren (b)	Forecast ke 18
alfa	1	Maret	Rp 26.131.324.832	1	Rp 26.131.324.832			
beta	0.67	April	Rp 32.858.462.674	2	Rp 32.858.462.674	Rp 32.858.462.674	Rp 6.727.137.842	
phi	0.05	Mei	Rp 36.574.246.756	3	Rp 36.574.246.756	Rp 36.574.246.756	Rp 2.625.951.589	
		Juni	Rp 44.975.149.464	4	Rp 44.975.149.464	Rp 44.975.149.464	Rp 5.712.081.504	Rp 49.618.030.042
		Juli	Rp 49.614.523.315	5	Rp 49.614.523.315	Rp 49.614.523.315	Rp 3.230.637.109	Rp 52.052.876.499
		Agustus	Rp 49.676.829.516	6	Rp 49.676.829.516	Rp 49.676.829.516	Rp 99.745.150	Rp 49.746.322.045
		September	Rp 49.837.200.480	7	Rp 49.837.200.480	Rp 49.837.200.480	Rp 109.928.699	Rp 49.907.405.611
		Oktober	Rp 50.535.677.086	8	Rp 50.535.677.086	Rp 50.535.677.086	Rp 472.982.795	Rp 50.810.283.341
		November	Rp 50.535.677.086	9	Rp 50.535.677.086	Rp 50.535.677.086	Rp 8.451.824	Rp 50.540.093.381
		Desember	Rp 50.746.497.979	10	Rp 50.746.497.979	Rp 50.746.497.979	Rp 142.318.325	Rp 50.812.600.175
		Januari	Rp 50.942.464.827	11	Rp 50.942.464.827	Rp 50.942.464.827	Rp 134.693.571	Rp 50.997.205.482
		Februari	Rp 51.058.673.197	12	Rp 51.058.673.197	Rp 51.058.673.197	Rp 80.772.106	Rp 51.086.810.193
		Maret	Rp 51.190.856.981	13	Rp 51.190.856.981	Rp 51.190.856.981	Rp 90.581.611	Rp 51.217.152.086
		April	Rp 52.249.072.088	14	Rp 52.249.072.088	Rp 52.249.072.088	Rp 715.227.126	Rp 52.415.170.360
		Mei	Rp 50.132.073.673	15	Rp 50.132.073.673	Rp 50.132.073.673	-Rp 1.414.819.642	Rp 52.251.297.773

Rumus Dasar

$$L_t = \alpha X_t + (1 - \alpha)(L_{t-1} + \phi b_{t-1})$$
$$b_t = \beta(L_t - L_{t-1}) + (1 - \beta)\phi b_{t-1}$$
$$F_{t+m} = L_t + (\phi^1 + \phi^2 + \dots + \phi^m)(b_t)m$$

$$\phi \cdot \frac{1 - \phi^m}{1 - \phi}$$

MAPE

2,1%

Logistic Growth Model Non Linear Square

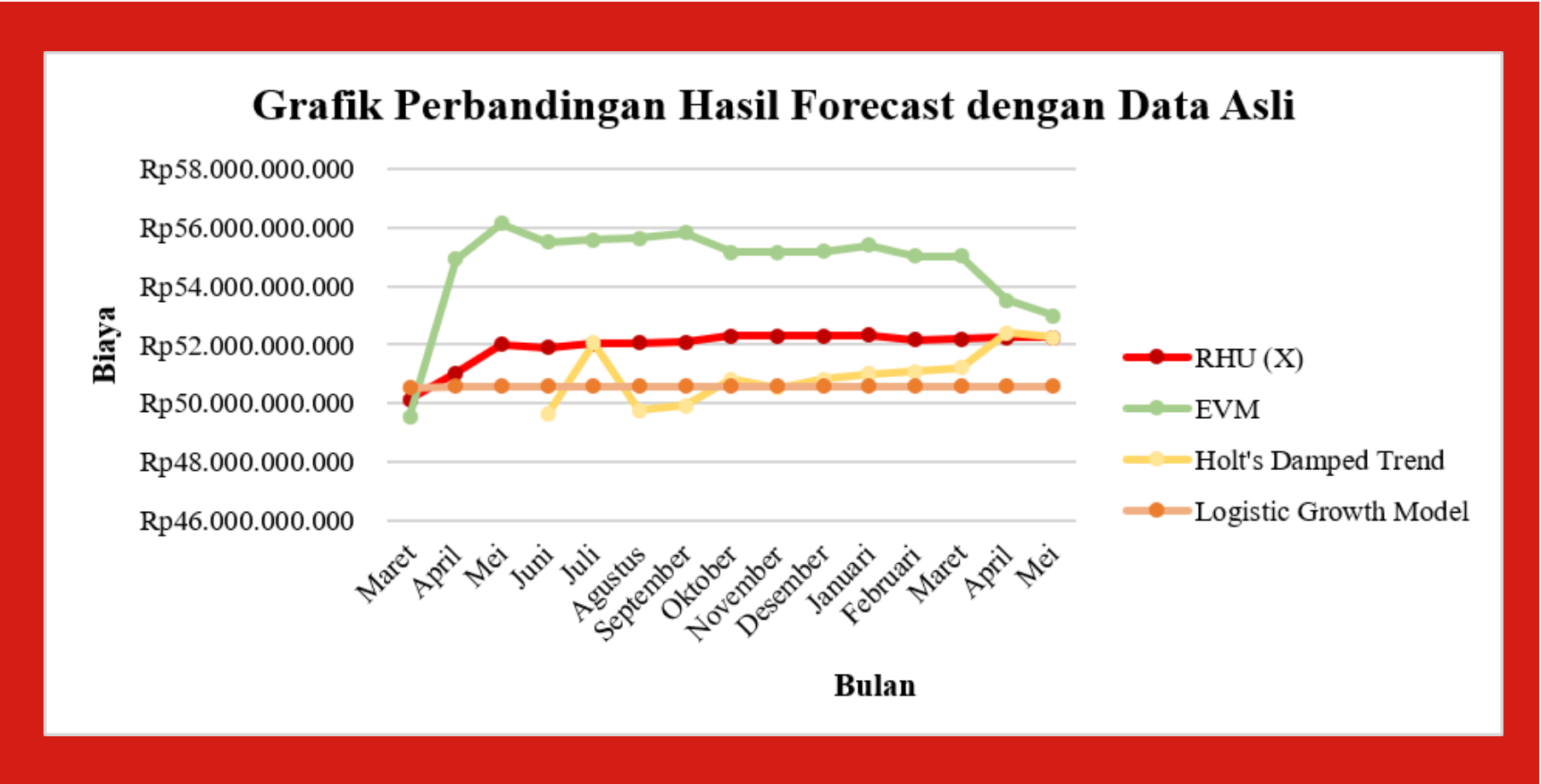
K	Rp 50.568.600.331	Rencana Budget	
r	59		
t	Bulan	X(t) -> P(0)	P(t)
1	Maret	Rp 26.131.324.832	Rp 50.566.513.163
2	April	Rp 32.858.462.674	Rp 50.567.397.379
3	Mei	Rp 36.574.246.756	Rp 50.567.746.338
4	Juni	Rp 44.975.149.464	Rp 50.568.322.750
5	Juli	Rp 49.614.523.315	Rp 50.568.557.411
6	Agustus	Rp 49.676.829.516	Rp 50.568.560.264
7	September	Rp 49.837.200.480	Rp 50.568.567.575
8	Oktober	Rp 50.535.677.086	Rp 50.568.598.877
9	November	Rp 50.535.677.086	Rp 50.568.598.877
10	Desember	Rp 50.746.497.979	Rp 50.568.608.155
11	Januari	Rp 50.942.464.827	Rp 50.568.616.711
12	Februari	Rp 51.058.673.197	Rp 50.568.621.753
13	Maret	Rp 51.190.856.981	Rp 50.568.627.461
14	April	Rp 52.249.072.088	Rp 50.568.672.116
15	Mei	Rp 52.249.072.088	Rp 50.568.672.116

Rumus Dasar

$$P(t) = \frac{P(0) e^{rt}}{1 + P(0) (e^{rt} - 1) / K}$$

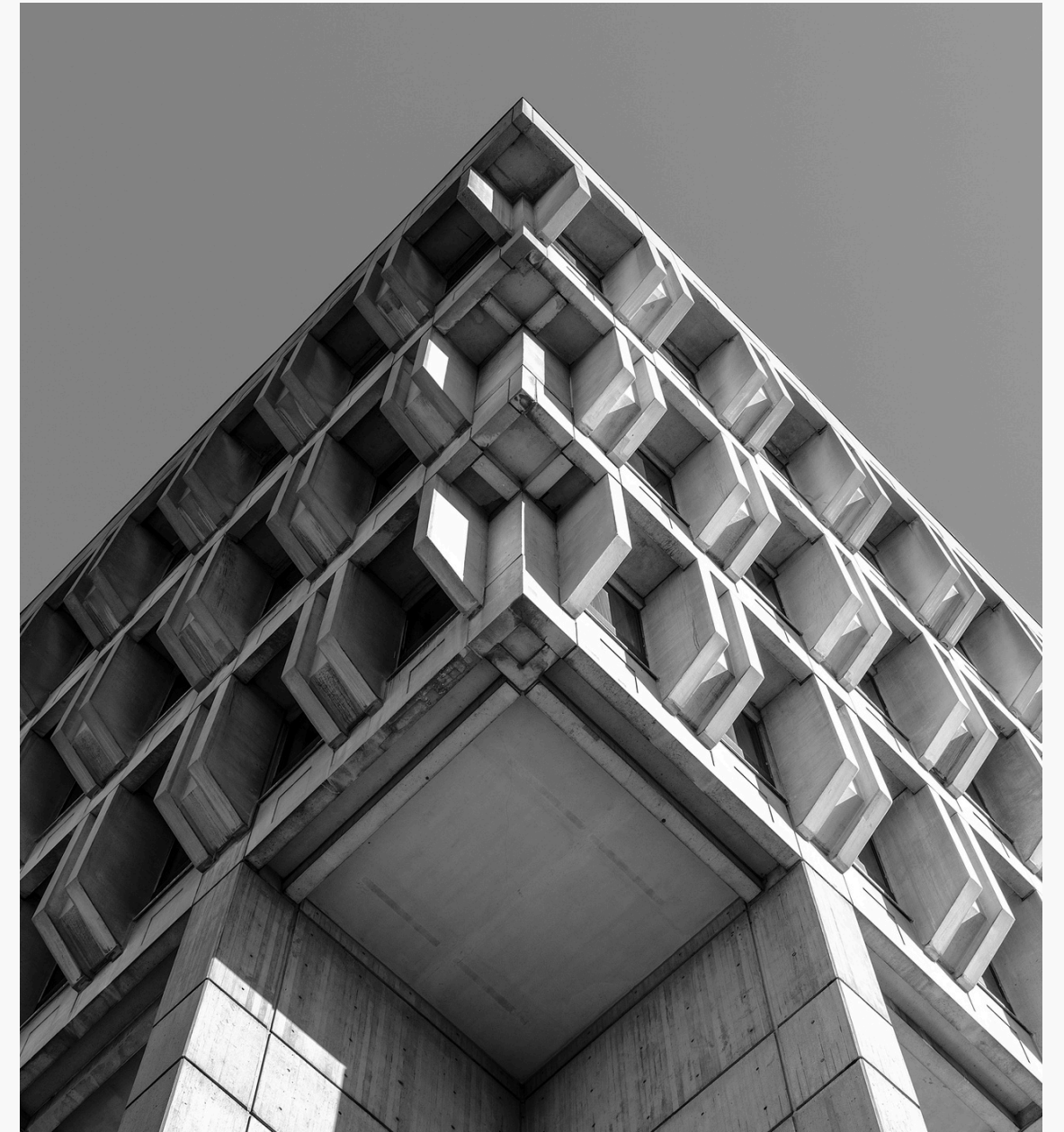
MAPE
2,8%

Periode (t)	EVM	Holt's Damped Trend	Logistic Growth Model
1	Rp 49.541.349.674		Rp 51.354.040.645
2	Rp 54.917.531.963		Rp 51.354.952.617
3	Rp 56.153.561.696		Rp 51.355.312.531
4	Rp 55.514.080.045	Rp 49.618.030.042	Rp 51.355.907.037
5	Rp 55.583.994.211	Rp 52.052.876.499	Rp 51.356.149.064
6	Rp 55.653.796.907	Rp 49.746.322.045	Rp 51.356.152.007
7	Rp 55.833.463.224	Rp 49.907.405.611	Rp 51.356.159.547
8	Rp 55.173.896.385	Rp 50.810.283.341	Rp 51.356.191.832
9	Rp 55.173.896.385	Rp 50.540.093.381	Rp 51.356.191.832
10	Rp 55.191.561.615	Rp 50.812.600.175	Rp 51.356.201.401
11	Rp 55.404.693.887	Rp 50.997.205.482	Rp 51.356.210.226
12	Rp 55.044.307.978	Rp 51.086.810.193	Rp 51.356.215.427
13	Rp 55.037.717.561	Rp 51.217.152.086	Rp 51.356.221.314
14	Rp 53.519.867.661	Rp 52.415.170.360	Rp 51.356.267.370
15	Rp 52.997.065.504	Rp 52.251.297.773	Rp 51.356.173.286



Insight

- Grafik EVM melonjak tinggi saat di awal , baru menurun secara perlahan
- Grafik Holt’s Damped Trend terlihat fluktuatif, cenderung di bawah grafik data asli (RHU)
- Grafik logistic growth model NLS paling stabil, tapi tidak bisa menangkap lonjakan yang mendadak



Kesimpulan

- Setiap metode atau model memiliki kelebihan dan kekurangannya masing – masing
- Holt's Damped Trend memiliki MAPE yang paling kecil → paling baik diantara dua metode lainnya
- Rekomendasi : mengkombinasikan antara metode Holt's Damped Trend dan Logistic Growth Model

THANKYOU